Local Conservation for the Hard-Sphere Boltzmann Equation

Exact Collision Measures, Weak Ram Moments,
Positive Paired Cubature, and Quantified Truncation

Dr. Muralidhara Subbarao
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www.integralresearch.net
Stony Brook University
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Abstract

The pinned OpenAI manuscript constructs two global entropy solutions of the periodic hard-sphere Boltzmann equation with the same datum and exact local mass, momentum, and energy balances. We examine its full mathematical proof route while separating that inherited claim from the local results proved here. The collision-plane coefficient, spherical collision mass, axial atoms, measurable compressed parameter maps, and singular velocity weight are retained exactly. A weak Ram formulation expands smooth observations, not rough sheet densities. We prove polynomially weighted observation-tail bounds, an invariant-quotient error estimate, and positive finite cubature on physical collision quadruples. Paired gains and losses cancel the five invariants for every node and every common truncation. An explicit angular error bound and exact quadratic stress contraction supplement a compact small-mass estimate with optimized exponent 2/3; its truncation errors and deteriorating constants remain visible. None of these finite constructions establishes compactness, entropy admissibility, branching, or nonuniqueness. Contribution assessment: C2/5, useful scoped weak-observation and truncation refinements are proved, while the global construction and its decisive concentration and branching arguments remain inherited.

1 Problem, evidence boundary, and source map

The primary object is OpenAI's October 5, 2026 preprint *Nonuniqueness with local conservation for the hard-sphere Boltzmann equation* [1]. The equation is posed on $\mathbb{T}^3_x \times \mathbb{R}^3_v$, with torus volume one:

$$DF = Q(F,F), \qquad D = \partial_t + v \cdot \nabla_x, \qquad F \geq 0.$$

The manuscript claims a bounded-velocity-support initial density with finite mass, energy, and absolute entropy integral, and two distinct global solutions satisfying the same initial trace. Its admissibility definition includes renormalized identities relative to a Maxwellian, entropy dissipation, weighted flux integrability, and all five *local* conservation laws. The resulting solutions are asserted to be strongly continuous in L^1, with separately integrable gains and losses carrying every fixed polynomial velocity weight on finite time intervals.

Reading that proof route is not a formal certification of its global theorem. The present additions concern exact collision measures and weak observations at a fixed time and position, with optional integration over time and space under explicit integrable bounds. They neither construct a second

solution nor show that a numerical collision rule converges to one. Applicability level L4 records relevance of the private methods; the realized contribution is assessed independently as C2.

Pinned module	Mathematical role and boundary
01–02	Admissibility, hard-sphere normalization, rotated periodic lattice, singular weight, and characteristic representatives.
03	Exact Carleman planes; arbitrary axial laws; Gaussian parameter sheets; normalized tails; hot gain and qualitative small-mass estimates.
04–06	Finite colored cutoff solutions, entropy, cold barriers, retirement geometry, seed cap, and uniform early continuation.
07–08	Global bath bootstrap, weighted collision-product uniform integrability, velocity averaging, strong traces, and local conservation.
09–10	Atomic two-beam dilation, cold–hot product passage, two axial gains, five-gain rank, and nonvanishing of cap limits.
11–12	Positive branching histories, finite dilated horizon, and separation from the dormant solution.

All twelve mathematical modules were read in full, in bounded displays; the exact line intervals and endpoint hashes are recorded separately. The build's main file, preamble, two figure sources, and bibliography were also read. Private reading was bounded to the passages specified in Section 10; their external included files were not silently counted as read.

2 Exact collision law and its domains

Use ordinary area measure $\mathrm{d}\sigma$ on S^2, of mass 4π. For incoming velocities p, ξ, define

$$g = p - \xi, \qquad m_0 = (p+\xi)/2, \qquad r = |g|/2, \qquad v = m_0 + r\sigma, \quad w = m_0 - r\sigma.$$

The notation m_0 here denotes a collision center, not the weight exponent introduced below. The single-output collision measure is

$$\kappa_{p,\xi}(\,\mathrm{d}\sigma) = \frac{|p-\xi|}{2}\,\mathrm{d}\sigma, \qquad \kappa_{p,\xi}(S^2) = 2\pi|p-\xi|. \tag{2.1}$$

It has zero mass on the coincident-input diagonal. If c is a positive finite Borel measure and h a nonnegative Lebesgue density, define the joint collision measure

$$\mathrm{d}\mu = c(\,\mathrm{d}p)h(\xi)\,\mathrm{d}\xi\,\frac{|p-\xi|}{2}\,\mathrm{d}\sigma. \tag{2.2}$$

Finite first moments of both inputs suffice for finite total mass. Higher weak observations below require the stated higher moments. An axial atom is an admissible input measure, not a smooth density in disguise. The gain is initially the pushforward of μ by v.

Proposition 2.1 (Carleman and spherical normalization; inherited)**.** *For the source's hard-sphere kernel $|(v-v_*)\cdot\omega|$, the symmetric gain P obeys*

$$P(c,h)(v) = 2\int c(\,\mathrm{d}p)\int \delta((v-p)\cdot(\xi-v))h(\xi)\,\mathrm{d}\xi. \tag{2.3}$$

For $p \neq v$, the inner delta denotes $\mathrm{d}A_\xi/|v-p|$ on the plane through v normal to $v-p$. The formula is a measure identity, and a density identity wherever the displayed disintegration is finite. Moreover,

$$\int P(c,h)(v)\,\mathrm{d}v = 2\pi\iint |p-\xi|c(\,\mathrm{d}p)h(\xi)\,\mathrm{d}\xi.$$

Proof. The relative-velocity reflection $g \mapsto g - 2(g \cdot \omega)\omega$ maps each hemisphere to the sphere of relative directions. On each hemisphere, the pushforward of $|g \cdot \omega|\, \mathrm{d}\omega$ is $|g|\, \mathrm{d}\sigma/4$. Both hemispheres give (2.1). The outgoing sphere has radius $|g|/2$, so its measure density relative to area is $2/|g|$. For fixed p, ξ, the function $f(v) = (v - p) \cdot (\xi - v)$ vanishes on this sphere and has $|\nabla_v f| = |g|$ there. Coarea therefore gives $2\delta(f(v))\, \mathrm{d}v$. Tonelli, followed by integration first in ξ, gives (2.3). The mass formula follows by integrating the sphere. For positive input measures all changes are justified as nonnegative measure identities; no differentiation of an atomic law is involved. The diagonal carries zero collision rate. □

The loss frequency is $\nu_c(v) = 2\pi \int |v - p| c(\, \mathrm{d}p)$. For densities the polarized cross operator is

$$\mathcal{C}(c, h) = 2P(c, h) - c\nu_h - h\nu_c.$$

For measure c, the same formula is interpreted weakly as a signed measure. It is not the source's branching operator $2P(B, h) - h\nu_B$: that operator omits the removal of the cold partner because the background is fixed. It need not conserve the hot mass. This distinction is essential when transporting conservation statements from the nonlinear equation to its dilated linear limit.

As a normalization check, for $c = \delta_0$ and $h(\xi) = e^{-\lambda|\xi|^2}$, $\lambda > 0$, the plane integral gives

$$P(\delta_0, h)(v) = \frac{2\pi}{\lambda|v|} e^{-\lambda|v|^2} \quad (v \neq 0), \qquad \int P(\delta_0, h) = \frac{4\pi^2}{\lambda^2}.$$

The last value is also $2\pi \int |\xi| e^{-\lambda|\xi|^2}\, \mathrm{d}\xi$. An incorrect probability normalization or a missing hemisphere changes this equality.

3 Positive sheets and the actual normalized kernel

Write $v = (\bar{v}, v_3)$ and fix

$$H(v) = e^{-|v|^2} \langle v \rangle^{-m} W(v), \qquad W(v) = 1 + |\bar{v}|^{-q}, \qquad 1 < q < 2, \quad m = 100.$$

All assertions involving division by H are essential assertions off the velocity axis. The axis has Lebesgue measure zero, while H is locally integrable and has every polynomial moment. The conditions $q > 1$ and $q < 2$ serve different purposes: integration along transverse lines and integration across the two-dimensional transverse plane.

Let $\mathcal{G}_d$ be the transverse Gaussian of density $(\pi d^2)^{-1} e^{-|\bar{p}|^2/d^2}$, with $\mathcal{G}_0 = \delta_0$. A source probability sheet is $\mathcal{G}_d \otimes \rho$, where $0 \leq d \leq 1/8$ and $\int e^{8a^2} \rho(\, \mathrm{d}a) \leq L$. Positive probability mixtures form $\mathcal{S}_L$; actual cold measures can be dominated by $A_0 c_0$, $c_0 \in \mathcal{S}_L$, with amplitude $A_0 \geq 0$. This amplitude is retained in every estimate.

The source defines

$$K_c(v;\, \mathrm{d}p\, \mathrm{d}\xi) = 2c(\, \mathrm{d}p)\delta\big((v - p) \cdot (\xi - v)\big) \frac{H(\xi)}{H(v)}\, \mathrm{d}\xi. \tag{3.1}$$

Its total mass is $P(c, H)(v)/H(v)$, not one. For the linear gain $2P(c, \cdot)$ the corresponding kernel is $2K_c$, whose plane coefficient is four. Here *normalized* means normalized by the output weight, not made into a probability kernel.

3.1 What is already proved in the source

The following estimates are inherited from its complete kernel module:

$$\operatorname*{ess\,sup}_{v} K_c(v;\mathrm{all}) \le CA_0, \tag{3.2}$$

$$K_c(v;R<S\le 2R) \le CA_0R^2e^{-c_*R^2}, \quad S=|p|+|\xi-v|, \quad R\ge 1, \tag{3.3}$$

$$K_c(v;S\le R,\ \xi_3\in I) \le C_RA_0\big(|I|+|I|^{2-q}\big), \quad 0<|I|\le 1. \tag{3.4}$$

Constants depend on q, m, L, and C_R also on R. The full mass tends uniformly to zero at large output velocity. No analogous uniform vanishing at the output axis is asserted.

The geometric inputs are not replaced by a matrix positivity test. A transverse line integral of the singular part is

$$a_q|n|^{-1}|A-Bs|^{1-q}, \qquad a_q=\int_{\mathbb{R}}(1+t^2)^{-q/2}\,\mathrm{d}t<\infty.$$

On an axial interval of length ℓ inside $[-R,R]$, its integral is bounded by $CR^{q-1}\ell^{2-q}|A|^{1-q}/|n|$, in the extended sense at $A=0$. Averaging transverse disk laws controls the singular angular factor because $q<2$; a transverse Gaussian is a positive mixture of such disk laws. This yields the source's two geometric terms $R\ell+R^{q-1}\ell^{2-q}$, before the bounded Gaussian and polynomial factors on $S\le R$ are inserted.

The shell decay uses the exact collision identity

$$k=\xi-v, \quad w=p+k, \quad |p|^2+|\xi|^2=|v|^2+|w|^2 \quad \text{when } (v-p)\cdot k=0. \tag{3.5}$$

Thus the Gaussian ratio becomes $e^{|p|^2-|w|^2}$. The polynomial ratio is bounded by a constant times $e^{|p|^2}$. Tilting the cold law by $e^{2|p|^2}$ leaves a uniformly controlled sheet law and a Gaussian tail, using the width bound and the axial exponential moment. If $S>R$, either $|p|>R/4$ or $|w|>R/2$. The remaining plane integral costs $O(R^2)$. This is the origin of (3.3), not an assumed Gaussian law for the whole collision.

3.2 Measurable compression and atomic laws

For a nonnegative parameter density

$$j(a)\le C_0\pi^{-3/2}e^{-|a|^2}, \qquad p(a)=(d\bar{a},\Phi(a_3)), \qquad |\Phi(s)|\le A+\delta|s|, \quad d,\delta\le 1/16,$$

the source proves domination by $C_0\mathcal{S}_{L_A}$, where $L_A=2e^{16A^2}$. The map Φ is only measurable. Indeed $e^{8\Phi(s)^2}\le e^{16A^2}e^{16\delta^2s^2}$; Gaussian integration controls this moment, including for the wider Gaussian used in the parameter tail. No inverse Jacobian is required. The restricted pushforward from $|a|>B$ satisfies

$$\|P(c_{|a|>B},H)/H\|_\infty \le CC_0e^{-B^2/2}, \tag{3.6}$$

because $\mathbf{1}_{|a|>B}e^{-|a|^2}\le e^{-B^2/2}e^{-|a|^2/2}$. Positive mixtures and measurable masks in $[0,1]$ preserve these dominations.

The smooth implicit map $p_3=\upsilon(z-tp_3)+da_3$, $\upsilon'\ge 0$, is one application, not the definition of the class: $\partial p_3/\partial a_3=d/(1+t\upsilon')$. Its compression may be arbitrarily strong. A constant Φ, $d=0$, gives an atom. Atoms, the exact factor two, the parameter tail, the ℓ^{2-q} interval modulus, the hot estimate $P(H,H)\le CH$ for $m>5$, and qualitative small-mass gain are all source results, not contributions of this report.

4 Paired weak moments and invariant cancellation

For a real observation ψ, define

$$\Delta\psi(p,\xi,\sigma) = \psi(v) + \psi(w) - \psi(p) - \psi(\xi), \qquad \mathcal{I} = \operatorname{span}\{1, v_1, v_2, v_3, |v|^2/2\}.$$

Assume absolute integrability of the four observations under μ. For instance, growth at most $\langle v\rangle^k$ is allowed if the two inputs have sufficient $(k+1)$-moments.

Proposition 4.1 (Exact weak polarization)**.** *With μ as in (2.2),*

$$\langle \mathcal{C}(c,h), \psi\rangle = \int \Delta\psi \, \mathrm{d}\mu. \tag{4.1}$$

For a density f, the diagonal operator obeys $\langle Q(f,f),\psi\rangle = \frac{1}{2}\int \Delta\psi \, \mathrm{d}\mu_{f,f}$. Every $I \in \mathcal{I}$ has $\Delta I = 0$ pointwise.

Proof. The v and w marginals of μ coincide under $\sigma \mapsto -\sigma$, and each equals $P(c,h)$. The incoming p marginal has weight $c(\,\mathrm{d}p)2\pi\int |p-\xi| h(\xi)\,\mathrm{d}\xi = c\nu_h$; the other equals $h\nu_c$. This proves (4.1). For $c = h = f$, polarization gives $\mathcal{C}(f,f) = 2Q(f,f)$. Finally $v + w = p + \xi$ and $|v|^2 + |w|^2 = |p|^2 + |\xi|^2$, proving all five cancellations. □

For any common measurable collision mask $0 \le \chi \le 1$, define $\mathcal{C}_\chi$ by $\int \chi\Delta\psi \, \mathrm{d}\mu$. It preserves the invariants even when χ is not invariant under exchange of the outgoing labels. It is a paired weak modification; without that exchange symmetry it must not be identified with twice a separately masked single-output gain. Similarly, any finite rule

$$\langle \mathcal{C}_{\mathrm{disc}}, \psi\rangle = \sum_{\ell=1}^{J} a_\ell[\psi(v_\ell) + \psi(w_\ell) - \psi(p_\ell) - \psi(\xi_\ell)], \qquad a_\ell \ge 0,$$

on physical elastic quadruples preserves all invariants exactly in exact arithmetic. Positivity refers to collision weights, not to the signed collision operator or to a time step's preservation of $F \ge 0$. An explicit time integrator still needs a separate positivity condition. Independently normalizing gain and loss rules would break this argument.

5 A weighted weak-observation tail certificate

Suppose $c \le A_0 c_0$, $c_0 \in \mathcal{S}_L$, and $0 \le h \le ZH$. Write $h = Hu$, $0 \le u \le Z$ almost everywhere. In collision-plane coordinates the *joint* measure, not just its output marginal, is

$$\mathrm{d}\mu = H(v)u(\xi)\,\mathrm{d}v\, K_c(v;\,\mathrm{d}p\,\mathrm{d}\xi), \qquad w = p + \xi - v. \tag{5.1}$$

This is the coarea identity of Proposition 2.1 applied to tests depending on all collision variables. Consequently it handles output-dependent masks without inventing a conditional probability.

Lemma 5.1 (Weighted normalized tails)**.** *For each real $k \ge 0$ there are $C_k, c_k > 0$ such that*

$$\operatorname*{ess\,sup}_v \int_{S>R} (1+S)^k K_c(v;\,\mathrm{d}p\,\mathrm{d}\xi) \le C_k A_0 R^{k+2} e^{-c_k R^2}, \qquad R \ge 1.$$

The same integral over all S is at most $C_k A_0$. For parameter sheets as in Section 3, restriction to $|a| > B$ gives an all-S bound $C_k C_0 e^{-B^2/2}$.

Proof. On $2^j R < S \le 2^{j+1}R$, apply (3.3) and bound $(1+S)^k \le C_k(2^jR)^k$. Summing gives

$$C_k A_0 R^{k+2} \sum_{j\ge 0} 2^{j(k+2)} e^{-c_* 4^j R^2} \le C_k' A_0 R^{k+2} e^{-(c_*/2)R^2}.$$

The last inequality follows by $R \ge 1$ and summability of $2^{j(k+2)}e^{-(c_*/2)4^j}$. The part $S \le 1$ uses (3.2). For the parameter tail, its measure is dominated by $2^{3/2}C_0 e^{-B^2/2}$ times a wider probability sheet, which still has uniform width and axial-moment bounds. Apply the all-S estimate to that sheet. This proves the weighted version, not merely the unweighted parameter-tail statement. □

For $k \ge 2$ define the invariant-quotient seminorm

$$\|\psi\|_{k/\mathcal{I}} = \inf_{I\in\mathcal{I}} \sup_{z\in\mathbb{R}^3} \frac{|\psi(z) - I(z)|}{\langle z\rangle^k}, \qquad H_k = \int \langle v\rangle^k H(v)\,\mathrm{d}v < \infty.$$

No minimizer in this finite-dimensional infimum is needed. Let $D_{R,M} = \{S \le R,\ |v| \le M\}$, $R, M \ge 1$. When a marked Gaussian parameter representation is available, additionally retain $|a| \le B$, obtaining $D_{B,R,M}$. The parameter is retained as part of the measure before pushforward; no measurable inverse of $p(a)$ is assumed.

Theorem 5.2 (Paired weak truncation error)**.** *Under the preceding sheet and hot-density hypotheses,*

$$|\langle \mathcal{C} - \mathcal{C}_{D_{R,M}}, \psi\rangle| \le C_k A_0 Z \|\psi\|_{k/\mathcal{I}} \left[H_k R^{k+2} e^{-c_k R^2} + \int_{|v|>M} \langle v\rangle^k H(v)\,\mathrm{d}v \right]. \tag{5.2}$$

For a parameter sheet of amplitude C_0, the $D_{B,R,M}$ version has the same bound with A_0 replaced by C_0, with an additional term $C_k C_0 Z H_k \|\psi\|_{k/\mathcal{I}} e^{-B^2/2}$. All masks use the same retained collision in both gains and both losses. The bounds vanish exactly for invariant observations.

Proof. Put $k_0 = \xi - v$. Then $|p| \le S$, $|w| = |p + k_0| \le S$, and $|\xi| \le |v| + S$. For $N_I = \sup_z |\psi(z) - I(z)|/\langle z\rangle^k$,

$$|\Delta\psi| = |\Delta(\psi - I)| \le C_k N_I \langle v\rangle^k (1+S)^k.$$

Insert (5.1), use $u \le Z$, and split the omitted set into $S > R$ and $|v| > M$. Lemma 5.1 gives (5.2). The marked parameter deletion is handled by the last statement of that lemma, and a union bound accounts for intersections. Infimize in I. For $I \in \mathcal{I}$, cancellation is pointwise before every bound or deletion. All interchanges are justified by the nonnegative dominating integrals just established. □

The remaining output tail is explicitly bounded by

$$\int_{|v|>M} \langle v\rangle^k H(v)\,\mathrm{d}v \le e^{-M^2/2} \int e^{-|v|^2/2} \langle v\rangle^{k-m} (1 + |\bar v|^{-q})\,\mathrm{d}v.$$

This finite constant depends on k, q, m; it does not require differentiating the axis singularity. These are continuum estimates conditional on the source sheet constants, not a finite algorithm for computing those constants. For a spacetime field, the pointwise bound can be integrated whenever $A_0(t,x)Z(t,x)$ times the indicated observation envelope is integrable. No uniform bound near the source's initial singular time is added silently.

6 Positive cubature on physical collisions

Push the retained collision measure onto $z = (p, \xi, v, w) \in (\mathbb{R}^3)^4$, and call it μ_D. Its support closure is compact: $S \le R$, $|v| \le M$ imply $|p|, |w| \le R$, $|\xi| \le M + R$. Every point of that closure is still elastic, since momentum and energy equalities define a closed set. Let $m_D = \mu_D(\mathbb{R}^{12})$. The zero-mass case is the empty rule.

Theorem 6.1 (Finite observation bank)**.** *Let $\psi_1, \dots, \psi_d$ be continuous on the retained velocity compact set. There exist at most $d+1$ physical collision quadruples $z_\ell \in \operatorname{supp} \mu_D$ and weights $a_\ell > 0$, with $\sum_\ell a_\ell = m_D$, such that*

$$\sum_\ell a_\ell \Delta\psi_j(z_\ell) = \int \Delta\psi_j \, \mathrm{d}\mu_D \quad (1 \le j \le d).$$

Every invariant is canceled by this same rule without adding a feature row. For observations in the polynomial growth class of Theorem 5.2, the total error relative to the full collision operator is bounded there.

Proof. For $m_D > 0$, map the compact support to $\Phi(z) = (\Delta\psi_1(z), \dots, \Delta\psi_d(z))$. The barycenter of its probability pushforward lies in the closed convex hull of its compact image: approximate the integral by simple sums, or separate a point outside that hull by a linear functional. Caratheodory reduction leaves at most $d+1$ points. Explicitly, any longer finite convex sum has an affine dependence; subtract the largest admissible multiple of that dependence to remove a weight without changing the sum or its mass. Compactness closes the resulting family of sums of at most $d+1$ points. Multiply the weights by m_D. Pointwise elasticity proves the invariant assertion. □

The convex argument is classical, consistent with the positive cubature theory of Bayer and Teichmann [5]; it is not claimed as a new existence theorem. Here it is applied to paired collision increments and combined with the explicit sheet-tail accounting. The nodes are in the closed velocity support, not necessarily in the pointwise image of a merely measurable parameterization. Recovering specific parameter preimages is not part of the theorem.

The theorem is existential. It does not promise that a prescribed grid admits such weights, or that finitely many samples of a measurable h determine its collision integrals. Changing h at the sampled points does not change its Lebesgue density class but may change a point-sampling algorithm. A usable implementation must obtain the collision weights from integrated cells, a specified atomic law, or a separately bounded approximation. If the actual finite rule has a verified bank residual e_j, its observation error is at most the corresponding tail bound plus $|e_j|$. This is an error identity, not an assertion that finite tests certify the needed continuum residual.

7 Weak Ram expansion: differentiate observations, not sheets

The private local-moment theorem [7] expands the acted-on function on the actual displacement segments and retains an integral remainder. Its valid transfer here is to the *test observation*. There is no hypothesis that h/H, the cold sheet, or a spatial stripe mask is differentiable. In particular no Taylor series of F is carried across the source's annular interfaces.

Set $z_0 = g/2$ and $z = r\sigma$. For $\psi \in C^{2n+2}$ on a neighborhood of the four segments $m_0 \pm sz, m_0 \pm sz_0$, $0 \le s \le 1$, Taylor's theorem through degree $2n+1$ gives

$$\Delta\psi = 2\sum_{j=1}^{n} \frac{D^{2j}\psi(m_0)[z^{2j}] - D^{2j}\psi(m_0)[z_0^{2j}]}{(2j)!} + \mathcal{R}_n. \tag{7.1}$$

If the multilinear norm of $D^{2n+2}\psi$ is bounded by M_{2n+2} on those segments, then

$$|\mathcal{R}_n| \le \frac{4M_{2n+2}r^{2n+2}}{(2n+2)!}. \tag{7.2}$$

The proof is direct: in each opposite pair all odd Taylor terms cancel; the constant terms cancel between the two pairs; the four integral remainders are each bounded by $M_{2n+2}r^{2n+2}/(2n+2)!$. Integration of this bound is legitimate only with the corresponding weighted collision moment. A high order does not by itself make a large-displacement remainder small.

Proposition 7.1 (A lower-regularity observation bound)**.** *Let $0 < \alpha \le 1$, and let $\nabla\psi$ have Holder seminorm L_α on the relevant convex velocity regions. Then*

$$|\Delta\psi| \le \frac{2^{1-\alpha}}{1+\alpha} L_\alpha |g|^{1+\alpha},$$

and

$$|\langle \mathcal{C}(c,h), \psi\rangle| \le \frac{2^{2-\alpha}\pi}{1+\alpha} L_\alpha \iint |p-\xi|^{2+\alpha} c(\mathrm{d}p) h(\xi)\,\mathrm{d}\xi.$$

These statements require the displayed moment to be finite and apply to atomic cold laws.

Proof. The first-order remainder at m_0 along displacement u is

$$\psi(m_0+u) - \psi(m_0) - \nabla\psi(m_0)\cdot u = \int_0^1 [\nabla\psi(m_0+su) - \nabla\psi(m_0)]\cdot u\,\mathrm{d}s.$$

Its absolute value is at most $L_\alpha|u|^{1+\alpha}/(1+\alpha)$. Sum the four remainders with $|u| = r$; the linear and constant terms cancel. This gives $4L_\alpha r^{1+\alpha}/(1+\alpha)$. Multiplication by the spherical mass $2\pi|g|$ gives the second assertion. □

Proposition 7.2 (Positive angular discretization)**.** *Partition S^2, up to area-zero boundaries, into measurable cells E_j with nodes $\sigma_j \in S^2$, and assume $|\sigma - \sigma_j| \le \delta_{\mathrm{ang}}$ on E_j. Use positive area weights $|E_j|$, and for each node keep both $m_0 + r\sigma_j$ and $m_0 - r\sigma_j$. Under Proposition 7.1's hypotheses, the error in the unmasked cross weak collision observation is at most*

$$\pi\delta_{\mathrm{ang}} L_\alpha \iint |p-\xi|^{2+\alpha} c(\mathrm{d}p) h(\xi)\,\mathrm{d}\xi. \tag{7.3}$$

For the diagonal equation the constant is divided by two.

Proof. Let $G(u) = \psi(m_0+u) + \psi(m_0-u)$. Then $\nabla G(u) = \nabla\psi(m_0+u) - \nabla\psi(m_0-u)$, so its norm on $|u| \le r$ is at most $L_\alpha(2r)^\alpha$. Integrating along the chord gives

$$|G(r\sigma) - G(r\sigma_j)| \le L_\alpha 2^\alpha r^{1+\alpha}|\sigma - \sigma_j| = \tfrac{1}{2}L_\alpha|g|^{1+\alpha}|\sigma-\sigma_j|.$$

The incoming terms are independent of σ. Sum over the cells, whose areas total 4π, and multiply by $|g|/2$. This proves (7.3). Every discrete increment still cancels the invariants, with no quadrature error for them. □

The proposition applies directly to the full sphere, or with exact weighted cell masses for a nonnegative angular mask. Replacing a rough mask by its point values is not covered: it introduces an additional measure error. Similarly, Theorem 5.2's output-dependent truncation is not silently differentiated in (7.3). Applying either proposition to $\psi - I$, $I \in \mathcal{I}$, can improve the observation bound while leaving the collision functional unchanged.

7.1 An exact stress formula

For a real symmetric matrix A and $\psi_A(v) = v^T A v$,

$$\langle \mathcal{C}(c,h), \psi_A \rangle = \pi \iint |g|^3 \left(\frac{\operatorname{tr} A}{3} - \widehat{g}^T A \widehat{g} \right) c(\,\mathrm{d}p) h(\xi)\,\mathrm{d}\xi. \tag{7.4}$$

At $g = 0$ the integrand is defined to be zero. Indeed $\Delta\psi_A = 2r^2(\sigma^T A\sigma - \widehat{g}^T A\widehat{g})$, and $\int \sigma_i\sigma_j\,\mathrm{d}\sigma = (4\pi/3)\delta_{ij}$. This proves (7.4), including its coefficient π. For $Q(f,f)$ it is $\pi/2$. The six directions $\pm e_i$, each of area weight $2\pi/3$, integrate this angular quadratic exactly; the isotropic matrix $A = aI$ yields zero. This does not assert closure of the evolving stress tensor: the relative speed factor and spatial dependence still depend on the full input law.

8 A two-scale small-mass modulus with explicit compact constants

The source already proves that, for $0 \le j \le BH$ and fixed B, $\|P(H,j)/H\|_\infty \le \varepsilon + C_{B,\varepsilon} \int j$. Its compact part contains a sharper explicit mass dependence. We record it without claiming a global power law after the deletions.

Fix $M \ge 1$, $L > M + 1$, $0 < \delta \le 1$, and put

$$K = \{|z| \le L,\ |\bar{z}| \ge \delta\}, \quad H_* = \sup_K H, \quad h_* = \inf_{|v| \le M} H(v) > 0, \quad B_K = BH_*, \quad m_j = \int j.$$

The infimum can be understood essentially, and $h_* \ge e^{-M^2}\langle M\rangle^{-m}$. Let $\omega_{\mathrm{out}}(M) = \operatorname{ess\,sup}_{|v|>M} P(H,H)(v)/H(v)$. The source proves $\omega_{\mathrm{out}}(M) \to 0$.

Theorem 8.1 (Optimized retained-part bound)**.** *For $m > 5$ there are constants depending on the weight such that*

$$\begin{aligned}\left\| \frac{P(H,j)}{H} \right\|_\infty &\le B\omega_{\mathrm{out}}(M) + CB(L-M)^{5-m} + BC_L(\delta^2 + \delta^{2-q}) \\ &\quad + \frac{2\pi L^2 H_*}{h_*} 3(\pi/2)^{1/3} B_K^{1/3} m_j^{2/3}.\end{aligned} \tag{8.1}$$

Here C_L may grow rapidly with L. No uniform dependence of C_L, H_*, h_*^{-1} on the deletion parameters is suppressed.

Proof. For $|v| > M$, positivity gives the first term. For $|v| \le M$, delete $|p| + |\xi| > L$. In the source's symmetrized hot–hot proof one first restricts to $|\xi| \ge |v|/2$, with factor two. The removed input region implies $|p| + |\xi - v| > L - M$. Its reduced normalized shell bound is CR^{5-m}; dyadic summation gives the second term. This uses the symmetric hot–hot majorant $BP(H,H)$, not a claimed unsymmetrized shell estimate.

On the remaining bounded input region delete $|\bar{p}| < \delta$ or $|\bar{\xi}| < \delta$. Symmetry handles either input. The transverse measure $(1 + |\bar{p}|^{-q})\mathbf{1}_{|\bar{p}|<\delta}\,\mathrm{d}\bar{p}$ is a positive disk mixture of mass $\pi\delta^2 + 2\pi\delta^{2-q}/(2-q)$. The axial interval has length at most $2L$. The geometric plane bound from Section 3, with bounded exponential factors at these inputs, gives the third term with a finite C_L.

Both retained inputs lie in K. Interchange inputs in the symmetric gain so that the rough j is outside the plane. A plane section of $\{|z| \le L\}$ has area at most πL^2; therefore

$$P(H\mathbf{1}_K, j\mathbf{1}_K)(v) \le 2\pi L^2 H_* \int \frac{j(p)\mathbf{1}_K(p)}{|v-p|}\,\mathrm{d}p.$$

For every $\rho > 0$, the last integral is at most

$$B_K \int_{|u|<\rho} |u|^{-1}\,\mathrm{d}u + \rho^{-1} m_j = 2\pi B_K \rho^2 + \frac{m_j}{\rho}.$$

If $B_K m_j > 0$, minimize at $\rho = (m_j/(4\pi B_K))^{1/3}$, obtaining $3(\pi/2)^{1/3} B_K^{1/3} m_j^{2/3}$. If either vanishes, the retained density vanishes and the bound follows directly. Divide by $H(v) \geq h_*$ and add the deletions. □

For fixed B, first choose M large, then L large, then δ small, and finally m_j small. This recovers the source's qualitative limit. At fixed truncations it supplies an explicit $m_j^{2/3}$ retained-part certificate. It does not establish $\|P(H,j)/H\|_\infty = O(m_j^{2/3})$ globally: shrinking the omitted tails changes all compact constants. Optimizing those changes would require quantitative output and axis moduli with tracked constants.

9 Where local conservation enters the global source argument

This section records the proof architecture actually read. It attributes the construction to the source and identifies which analytic assertions are not consequences of Sections 4–8.

9.1 Finite colors, cold barriers, and global continuation

After rotating the torus lattice so an irrational direction becomes axial, the source uses scales $s_j = s_0 e^{-j\eta}$, stripe periods s_j^2, thermal widths $d_s = d_0 s^{10}$, and retirement times $L_s = s^{-1/2}$. Each sign occupies one quarter of an uncut stripe period; an intersected annulus does not have an exact quarter-area formula. Jet mass is $O(s^2)$ and entropy is $O(s^2(1 + |\log s|))$, despite amplitude of order s^{-31}. Finite cutoff solutions exist in a weighted essential-supremum norm in free coordinates. For cutoff factor $\min\{1, b/|p-\xi|\}$, the same factor is kept in gain and loss; it is compatible with pre/post collision exchange.

Cold colors c_i, bath u, and hot part h satisfy

$$(D + \nu_F^b)c_i = P^b(c_i, c_i), \qquad (D + \nu_F^b)u = P^b(u,u),$$

$$(D + \nu_F^b)h = P^b(h,h) + 2P^b(c+u,h) + 2P^b(c,u) + \sum_{i \neq i'} P^b(c_i, c_{i'}).$$

The last sum is ordered. Summing gives the total Boltzmann equation. At retirement a cold color is added to the regular part, so the total density and its entropy do not jump.

For a monotone axial profile $U = v(z - tU)$,

$$U_z = \frac{v'}{1 + tv'} \geq 0, \qquad DU = (v_3 - U)U_z, \qquad D|v - Ue_3|^2 = -2U_z(v_3 - U)^2.$$

This favorable sign is indispensable in the cold barrier. Spatial masks are not differentiated. Completing the distance squares along flights gives the source's supersolution and

$$c_i \leq f_i + \epsilon_s m_i, \qquad \epsilon_s = C_* L_s d_s / s.$$

The exact pushforward coefficient of a free core is

$$\frac{KS(\gamma(z - tp_3)/s)}{s + t\alpha\gamma S(\gamma(z - tp_3)/s)}\, J(a)\,\mathrm{d}a, \quad S = \mathrm{sech}^2,$$

with the original masks retained. Its sheet amplitude is $C/(t+s)$; the cold remainder amplitude is $CL_s d_s/s^2 = O(s^{15/2})$. Distinct active tubes are disjoint using the lattice estimate $|\bar{n}| \geq c/(1+|n|)$. Polynomial losses from copies and scales are absorbed by nonmain Gaussian tails.

The source's bath is initially a Maxwellian with a spatial hole and velocity cutoff. The early hot cap is $\|h(t)\|_H \leq T^p(t+s_N)$, attained for a selected seed parameter. Its source is arbitrarily small to algebraic order near zero because the cold tubes lie in the bath hole. Reaching a fixed bath time t_1 also uses integrated smallness along free paths, not just a pointwise bound with an uncontrolled hole-dependent constant.

For late times the regular component w is bootstrapped below BH. Small *spatially integrated* entropy distance from the Maxwellian does not mean pointwise smallness. The source first proves a free-flight averaging bound: for $\tau \geq \rho > 0$,

$$\int_V A(s, x-\tau v)\,\mathrm{d}v \leq C_{V,\rho}\|A(s)\|_{L^1_x}$$

on the rotated torus. Truncating H, deleting short time differences, and retaining exponential damping converts this into small velocity averages uniformly over late endpoints. Two separated velocity balls then give $\nu_F^b \geq \kappa > 0$, with κ fixed before B. This yields exponentially decaying cold colors and summable retirement increments $Cs^{-C}e^{-\kappa(s^{-1/2}-t_1)}$.

The positive excess $j = (w-\mu)_+$ has small mass after the flight averaging argument. The small-mass gain lemma then gives $P^b(w,w) \leq C_5 H$ with C_5 chosen independently of B. Gronwall with the integrable coefficient $e^{-\kappa(t-t_1)}$ produces a strict bootstrap improvement. The source explicitly chooses the branching parameters first, then damping and B, entropy tolerance, hole and bath cutoff, hot tolerance, and final scale/horizon. Theorem 8.1 refines one retained integral; it does not replace this order of choices or give numerical values for the global continuation parameters.

9.2 Uniform integrability and actual invariant balances

For $R_{k,N}(t,x) = \int \langle v\rangle^k F_N(t,x,v)\,\mathrm{d}v$, the source obtains

$$R_{k,N} \leq C_{k,\tau}\left(1 + \sum_j \frac{\mathbf{1}_{\mathcal{A}_{s_j}(t)}}{t+s_j}\right), \qquad |\mathcal{A}_s(t)| \leq C_\tau s^2.$$

The active sets are disjoint, so the square has no products between distinct singular summands, and

$$\int_0^\tau \frac{s^2}{(t+s)^2}\,\mathrm{d}t = s - \frac{s^2}{\tau+s} \leq s.$$

Discarding small scales costs $C\sum_{s_j<\varepsilon}(s_j+s_j^2) \to 0$. Outside them, a fixed integrable velocity majorant controls the remaining finite scales and the regular part. This proves weighted uniform integrability of same-position products $F_N(t,x,v)F_N(t,x,v_*)$, not merely of their separate factors. Collision moments satisfy $\int \langle v\rangle^k Q_N^{\pm}\,\mathrm{d}v \leq C_k R^2_{k+1,N}$.

The source proves the needed L^1 velocity averaging statement by localizing the transport equation, applying the damped inverse

$$Rr(t,x,v) = \int_0^\infty e^{-s} r(t-s, x-sv, v)\,\mathrm{d}s,$$

truncating uniformly integrable sources to bounded L^2 functions, and using decay of $\int |\psi(v)|^2/[1+(\sigma+k\cdot v)^2]\,\mathrm{d}v$ at high frequency. Strong convergence of velocity averages plus uniform integrability

identifies the pair product by separated velocity tests and Vitali. Pre/post change of variables and weighted tails then identify $Q_N^{\pm} \rightharpoonup Q^{\pm}(F,F)$ separately.

With the unrenormalized source in L^1, free coordinates give

$$F(t,x+tv,v) = F_0(x,v) + \int_0^t Q(F,F)(s,x+sv,v)\,\mathrm{d}s$$

as a Bochner L^1 identity. This supplies strong traces and the pathwise renormalized chain rule. For $\psi \in \{1, v_1, v_2, v_3, |v|^2/2\}$, the actual local balance is

$$\int_0^\infty \int F\psi(\partial_t\varphi + v\cdot\nabla_x\varphi)\,\mathrm{d}x\,\mathrm{d}v\,\mathrm{d}t + \int F_0\psi\varphi(0)\,\mathrm{d}x\,\mathrm{d}v = 0.$$

Energy flux requires order-three velocity integrability. Exact node cancellation alone would not pass this identity through a weak limit without the weighted tightness and product identification just described. Likewise, entropy dissipation uses weak lower semicontinuity of $(A-B)\log(A/B)$ applied to primed and unprimed pair products; it is not certified by a finite set of invariant moments.

9.3 Nonvanishing, branching, and separation

The cap dilation has

$$r_N = t_{*,N} + s_N, \quad l_N = s_N/r_N, \quad U_N = T_N/r_N, \quad h_{r_N}(t,x,v) = \frac{h_N(r_N t, r_N x, v)}{T_N^p r_N}.$$

Its bound is $0 \le h_{r_N} \le (t+l_N)H$, with attained weighted norm one at $t = 1 - l_N$. Free-core survival, the exact axial Jacobian, and stripe averaging give a limiting two-atom background on each mesh annulus:

$$B = \sum_{\sigma=\pm1} \frac{(K/4)S(\gamma(z-tu_{\varrho,\sigma})/\varrho)}{\varrho + t\alpha\gamma S(\gamma(z-tu_{\varrho,\sigma})/\varrho)}\delta_{u_{\varrho,\sigma}e_3}, \qquad u_{\varrho,\sigma} = \sigma + \alpha\tanh\frac{\gamma(z-tu_{\varrho,\sigma})}{\varrho}.$$

The limit solves $Dg = 2P(B,g) - g\nu_B$. Weak-* cold averages are paired with strongly convergent hot averages; the cold factors do not converge as ordinary uniformly integrable velocity densities. The bath, hot quadratic term, and initial source scale respectively as $O(r_N)$, $O(T_N^p r_N^2)$, and $O(T_N^{M-1-p})$, for $M > p+1$.

Weak convergence does not preserve the attained essential supremum. The source therefore uses two gains to exclude an axial endpoint, with the axial interval modulus and characteristic matching, and five gains to regularize away from the axis. The redundant plane identity is

$$\int_{n^\perp} F(k)\frac{\mathrm{d}A(k)}{|n|} = \int_{\mathbb{R}^3} F(n\times L)|n|\zeta(L\cdot n/|n|)\,\mathrm{d}L, \qquad \int\zeta = 1.$$

Its factor $|n|$ is required because the transverse cross-product map has area Jacobian $|n|^2$. The five-gain map to (t_5, x_5, v_5) has rank seven almost everywhere on the analytic positive-time domain. An equal-time witness supplies a nonzero minor; analytic continuation, followed by a uniform local inverse-coordinate argument, handles measurable masks after deletions. This is a continuum rank and pullback proof, not a finite rank sample. The upper endpoint $U = 1$, $l = 1$, and $U = \infty$ require the separate cases treated in the source.

For the branching argument, the single-child law is a probability on the diameter sphere but its collision rate remains $2\pi|v-w|$. The source's reference medium is $\delta_{e_3} + \delta_{-e_3}$; retaining speeds in $|v| \le 5$ has probability at least $25/26$. The retained Markov rate is $2p(v)\nu_0(v)$, and its positive potential is

$(2p(v)-1)\nu_0(v) \geq 48\pi/13$. Three collisions yield a volume minorization. Mixing and bounded-speed histories supply a robust displacement block of mass multiplication greater than three, even for a measurably perturbed medium. The moving spatial boxes grow like t^3, while the cap adds one power of time. Choosing $(1+a_{\mathrm{br}})^4 < 3$ forces a finite dilated horizon. This argument deliberately has positive hot mass production; imposing the full polarized invariant cancellation on it would change the model.

Finally the dormant solution vanishes under horizon dilation off the velocity axis. If it equaled cap limits at arbitrarily small horizons, a diagonal choice of finite approximations would also vanish there. Nonvanishing and the finite horizon give $r_N/T_N \to \rho = 1/U > 0$. Comparing the two spacetime dilations costs ρ_N^{-5}: four powers from the spacetime Jacobian and one from the amplitude. It would force the nonzero cap limit to vanish off the axis, hence everywhere since it is a density dominated by H. These are inherited global dependencies. None follows from the finite bank or stress identities in this report.

10 Private proof lineage and limits of the transfer

The following confidential originals were consulted locally, not submitted to public search. Their contribution here is methodological and mathematical, with actual proof passages rather than title matching.

TR813, revised version 2. The full 278-line file [6] was read. Its finite-recovery proposition at lines 106–123 proves the weighted pseudoinverse error bound under full column rank and explains the null-space obstruction. Its layer discussion at lines 159–169 explicitly separates smooth panels, conservation modes, and interface errors. Neither proof supplies a coercive inverse for the present singular collision operator. We transfer the insistence on the actual domain and weak residual, not its unrelated synthetic viscous-shock test or its generic inverse template.

Local moments chapter, revision 2. Lines 1–235 of the 446-line original [7] were read. This includes the complete local representation and integral-remainder proof (26–63), source/output normalization distinction (163–184), and complete degree-closure proof (194–225). External files included at lines 92 and 161 were not read as part of this operation. The forward expansion needs regularity of the acted-on function along segments, not smoothness of the kernel label. That distinction leads to (7.1). Polynomial exactness does not imply closure or invertibility. The source's sheet-dependent moments are not asserted to form a nilpotent finite matrix.

Core pattern 16. Lines 99–370 of the 939-line original [8] were read, including the common-measure mixture identities, positive covariance proof, Gaussian contraction proof, finite-moment limitations, and complete remainder theorem. The identity $a^T \operatorname{Cov}(\phi)a = \mathbb{E}|a\cdot(\phi-\mathbb{E}\phi)|^2 \geq 0$ presupposes an actual positive law. A positive semidefinite finite table does not produce the collision-plane measure with its specified support. Accordingly Theorem 6.1 starts with μ_D, whose mass is retained explicitly. Normalizing it to μ_D/m_D is allowed only when $m_D > 0$, followed by multiplication by m_D. No probabilistic normalization is imposed on K_c, and no generic moment inversion is used to infer a Boltzmann solution.

11 Current primary literature and source status

The primary-source refresh was conducted on October 8, 2026 using public titles and public URLs only. DiPerna and Lions' publisher record [2] describes global existence and weak stability from physical

integral bounds, with velocity averaging and nonlinear normalization. That background does not identify the same-data nonuniqueness claim in the pinned manuscript.

Chen, Guo, Shen, and Zhang's current arXiv version [3] is v3, dated September 13, 2026, following the May 31 submission. Its Theorem 1.1 gives loss of uniform continuity for $0 \le s_0 < 1$: initial data are close in a Gaussian-weighted $L^2_v H^{s_0}_x$ norm but their solutions separate in the unweighted norm at shrinking positive times. It concerns the whole-space problem and nearby, not identical, initial data. It is neither the pinned periodic local-conservation theorem nor evidence that finite moments establish nonuniqueness.

Duan, Huang, Wang, and Yang [4] prove existence and uniqueness for a class with bounded velocity-weighted L^∞ amplitude and suitable smallness of an $L^1_x L^\infty_v$ norm and defect quantities. The pinned construction's unbounded shrinking jets cannot be placed in that class merely because their integrated entropy is small. Bayer and Teichmann [5] supply primary positive-cubature context; the compact convex proof here is included to expose exactly what it does and does not guarantee.

The refresh read the Chen et al. abstract and introductory material through the displayed construction outline, including Theorem 1.1; it did not independently audit their full proof. For Duan et al. and DiPerna–Lions, the primary abstract/metadata records were inspected, not full proofs. The six-page Bayer–Teichmann text was displayed and read, including its convex argument and measurable-feature corollary. The local pinned OpenAI mathematical sources were read in full. These are different scopes.

The live public GitHub directory request failed in the browsing tool. The provenance sidecar records byte comparison of the current local text with the authenticated October 8 cached Git trees, including the captured base and target commits. That is a dated source-identity check, not a new live-branch fetch, a publication-status guarantee, or mathematical validation. Endpoint pre-read and post-read hashes delimit the literal reading; they do not prove continuous absence of changes between those endpoints.

12 Benefits, costs, and concrete further work

Benefits. The paired formulation keeps collision geometry, plane mass, and all invariants in one object. Atomic axial limits and measurable compression remain admissible without fictitious sheet derivatives. The weighted tail estimate controls unbounded observations rather than just deleted probability. The invariant quotient avoids charging an error for a part of the observation that is annihilated exactly. The compact small-mass formula exposes the 2/3 power and every deletion constant. The stress identity provides a simple exact angular fixture tied to the actual hard-sphere relative-speed factor.

Costs and limitations. The bounds depend on sheet amplitudes, exponential axial moments, and the singular exponent $1 < q < 2$. Constants deteriorate toward those endpoints; the proof does not extend to $q = 1$ or $q = 2$. Cubature existence is not a stable algorithm, and a feature bank may be ill-conditioned or uninformative about omitted observations. An atomic rule has a different density class from the continuum solution and cannot be inserted unchanged into the source's entropy compactness proof. Collision conservation does not guarantee entropy dissipation or positivity after time stepping. Untracked interval, temporal, spatial, and law-approximation errors remain outside the local certificates. No speedup, benchmark improvement, or finite verification of a continuum theorem is claimed.

Constructive quadrature. A useful next step is integrated-cell approximation of the incoming measure with certified mass and weighted-tail bounds, followed by positive compression of paired

increments. The real obstacle is control of collision-measure approximation for measurable densities and singular planes, not the size of a formal moment matrix. A fixed grid may not contain any adequate physical elastic quadruples; grid enrichment and residual bounds must be part of the method. One should separately verify positivity and entropy behavior of any subsequent time integrator.

Quantitative global modulus. To convert (8.1) into a single explicit small-mass modulus, track ω_{out}, the bounded-input axis constants, and their dependence on L, δ, M, then optimize jointly. A global power $m_j^{2/3}$ cannot be read off from the compact Coulomb integral. The global bath bootstrap would additionally need all tolerance choices propagated through velocity averaging and retirement, in their stated order.

Broader kernels and observations. Angularly anisotropic cutoff kernels can retain paired invariants by using their actual positive angular measure and mass. The current Carleman coefficient and Gaussian shell constants would have to be rederived, not reused unchanged. Non-cutoff kernels require control of infinite angular mass using cancellation before integration; the finite positive measure argument is then insufficient. Larger observation banks can incorporate fourth-order stress or entropy proxies, but nonsmooth or unbounded tests need their own function-weighted tails and local regularity. An entropy proxy is not the entropy inequality.

13 Contribution and reproducible evidence

The realized contribution is C2: Theorem 5.2 gives a source-specific weighted weak-observation truncation bound with exact invariant cancellation; Theorem 6.1 applies classical positive compression to the physical paired collision law; Propositions 7.1–7.2 and (7.4) give observation-only residual and angular formulas; Theorem 8.1 quantifies the compact part of the source's small-mass proof. These are useful local refinements, not a new nonuniqueness theorem or a historically novel cubature principle.

The reproducible check sidecar binds its script and results to the final TeX hash. It records exact rational collision identities, normalization and stress fixtures, Taylor and Holder constants, small-mass optimization, and bounded numerical checks of the stated tail accounting. Test outcomes are reported there, not promoted to a universal proof. The read ledger records actual source/private intervals and hashes. Native compilation dispatch, physical compilation logs, page renders, all-page visual inspection, and the final contribution assessment are separate evidence. The freeze manifest binds only final deliverables, necessary evidence, and their actual inputs; it does not modify the shared catalogue.

14 Conclusion

For the pinned hard-sphere construction the correct local object is a positive collision measure with its physical rate, not an arbitrary normalized moment table. Smooth observations may be expanded with explicit remainder while the axial law remains atomic or measurable. Paired finite collision rules preserve the five invariants exactly, and the source's normalized sheet tails give rigorous bounds for omitted weak observations. The optimized compact small-mass estimate is useful only with its accompanying deletions. Weighted compactness, entropy, nonvanishing, branching, and separation remain distinct continuum arguments; they are not supplied by these local calculations.

References

[1] OpenAI. *Nonuniqueness with local conservation for the hard-sphere Boltzmann equation*. October 5, 2026. Pinned mathematical source, twelve modules. Pinned public repository directory.

[2] R. J. DiPerna and P.-L. Lions. *On the Cauchy problem for Boltzmann equations: Global existence and weak stability*. Annals of Mathematics 130 (1989), 321–366. Publisher record.

[3] X. Chen, Y. Guo, S. Shen, and Z. Zhang. *Sharp H^s_x Ill-posedness of the Hard-sphere Boltzmann Equation*. arXiv:2606.01331v3, September 13, 2026.

[4] R. Duan, F. Huang, Y. Wang, and T. Yang. *Global Well-Posedness of the Boltzmann Equation with Large Amplitude Initial Data*. Archive for Rational Mechanics and Analysis 225 (2017), 375–424. Primary preprint.

[5] C. Bayer and J. Teichmann. *The proof of Tchakaloff's Theorem*. arXiv:math/0502473v2, 2005.

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[7] M. Subbarao. *Local Moments and Ram Differential Operators*. Ram Calculus, Volume 1, revised chapter 3. Confidential local original; lines 1–235 of 446, excluding external included files.

[8] M. Subbarao. *Joint-Moment Bayesian Neural Inference and Inverse Uncertainty Propagation*. Core pattern 16 technical report. Confidential local original; lines 99–370 of 939.